

HealthSync — Deployment Guide

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1. Prerequisites

Hardware Requirements

Component	Minimum	Recommended
CPU	2 cores	4+ cores
RAM	4 GB	8+ GB
Storage	20 GB SSD	100 GB SSD
Network	100 Mbps	1 Gbps

Software Requirements

Software	Version	Purpose
Node.js	20.x LTS or 22.x	Runtime
npm	10.x+	Package manager
PostgreSQL	15+	Production database
Nginx	1.24+	Reverse proxy
Git	2.40+	Source control
OpenSSL	3.x	TLS certificates

Network Requirements

- Port 443 (HTTPS) open to hospital network
- Port 5432 (PostgreSQL) internal only

- Port 3000 (Node.js) internal only — behind reverse proxy
 - DNS record pointing to server IP
-

2. Environment Setup

2.1 Clone the Repository

```
git clone <repository-url> /opt/healthsync  
cd /opt/healthsync
```

2.2 Install Dependencies

```
npm ci --production=false
```

2.3 Configure Environment Variables

Create `/opt/healthsync/.env` :

```
# Database – PostgreSQL connection string  
DATABASE_URL="postgresql://healthsync_user:STRONG_PASSWORD_HERE@localhost:5432/healthsync  
schema=public"  
  
# JWT Secret – MUST be at least 32 characters, cryptographically random  
# Generate with: openssl rand -base64 48  
JWT_SECRET="REPLACE_WITH_CRYPTOGRAPHICALLY_RANDOM_SECRET_MIN_32_CHARS"  
  
# Runtime  
NODE_ENV="production"  
PORT=3000
```

CRITICAL: The JWT_SECRET must be:

- At least 32 characters long
- Cryptographically random (use `openssl rand -base64 48`)
- Not contain the string "dev-secret" (enforced at startup)
- Unique per environment (do not reuse between staging and production)

2.4 Secure the .env File

```
chmod 600 /opt/healthsync/.env  
chown healthsync:healthsync /opt/healthsync/.env
```

3. Database Configuration

3.1 Install PostgreSQL

```
# Ubuntu/Debian
sudo apt update && sudo apt install postgresql-15

# Start and enable
sudo systemctl enable postgresql
sudo systemctl start postgresql
```

3.2 Create Database and User

```
sudo -u postgres psql <<SQL
CREATE USER healthsync_user WITH PASSWORD 'STRONG_PASSWORD_HERE';
CREATE DATABASE healthsync_db OWNER healthsync_user;
GRANT ALL PRIVILEGES ON DATABASE healthsync_db TO healthsync_user;
SQL
```

3.3 Migrate from SQLite to PostgreSQL

Update `prisma/schema.prisma` :

```
datasource db {
  provider = "postgresql"
}
```

Run migrations:

```
npx prisma migrate deploy
```

3.4 Seed Initial Data

```
npx prisma db seed
```

This creates the default admin account. **Change the admin password immediately after first login.**

3.5 PostgreSQL Security Hardening

Edit `pg_hba.conf` :

```
# Reject all remote connections except from the app server
host healthsync_db healthsync_user 127.0.0.1/32 scram-sha-256
host all all 0.0.0.0/0 reject
```

Edit `postgresql.conf` :

```
listen_addresses = 'localhost'
ssl = on
ssl_cert_file = '/etc/ssl/certs/server.crt'
ssl_key_file = '/etc/ssl/private/server.key'
log_connections = on
```

```
log_disconnections = on
log_statement = 'mod'
```

4. Build and Deploy

4.1 Build the Application

```
cd /opt/healthsync
npm run build
```

Verify the build succeeds with no errors.

4.2 Create a System Service

Create `/etc/systemd/system/healthsync.service` :

```
[Unit]
Description=HealthSync Hospital Management System
After=network.target postgresql.service
Requires=postgresql.service

[Service]
Type=simple
User=healthsync
Group=healthsync
WorkingDirectory=/opt/healthsync
ExecStart=/usr/bin/node node_modules/.bin/next start -p 3000
Restart=always
RestartSec=5
Environment=NODE_ENV=production
EnvironmentFile=/opt/healthsync/.env

# Security hardening
NoNewPrivileges=yes
ProtectSystem=strict
ProtectHome=yes
ReadWritePaths=/opt/healthsync
PrivateTmp=yes

[Install]
WantedBy=multi-user.target
```

4.3 Start the Service

```
sudo systemctl daemon-reload
sudo systemctl enable healthsync
sudo systemctl start healthsync

# Verify it's running
```

```
sudo systemctl status healthsync
curl -s http://localhost:3000 | head -5
```

5. Reverse Proxy Configuration

5.1 Nginx Configuration

Create `/etc/nginx/sites-available/healthsync` :

```
upstream healthsync_backend {
    server 127.0.0.1:3000;
    keepalive 64;
}

server {
    listen 80;
    server_name healthsync.hospital.local;

    # Redirect all HTTP to HTTPS
    return 301 https://$server_name$request_uri;
}

server {
    listen 443 ssl http2;
    server_name healthsync.hospital.local;

    # SSL certificates
    ssl_certificate      /etc/ssl/certs/healthsync.crt;
    ssl_certificate_key  /etc/ssl/private/healthsync.key;
    ssl_protocols        TLSv1.2 TLSv1.3;
    ssl_ciphers           HIGH:!aNULL:!MD5:!3DES;
    ssl_prefer_server_ciphers on;
    ssl_session_cache    shared:SSL:10m;
    ssl_session_timeout  10m;

    # Security headers (supplementing application-level middleware)
    add_header X-Content-Type-Options "nosniff" always;
    add_header X-Frame-Options "DENY" always;
    add_header Strict-Transport-Security "max-age=31536000; includeSubDomains;
    preload" always;

    # Request limits
    client_max_body_size 10M;
    client_body_timeout 30s;
    client_header_timeout 30s;

    # Rate limiting zone (defined in nginx.conf http block)
    # limit_req_zone $binary_remote_addr zone=login:10m rate=5r/m;

    location / {
        proxy_pass http://healthsync_backend;
```

```

    proxy_http_version 1.1;
    proxy_set_header Upgrade $http_upgrade;
    proxy_set_header Connection 'upgrade';
    proxy_set_header Host $host;
    proxy_set_header X-Real-IP $remote_addr;
    proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
    proxy_set_header X-Forwarded-Proto $scheme;
    proxy_cache_bypass $http_upgrade;
    proxy_read_timeout 60s;
}

# Static assets – long cache
location /_next/static/ {
    proxy_pass http://healthsync_backend;
    expires 365d;
    add_header Cache-Control "public, immutable";
}

# Health check endpoint
location /api/health {
    proxy_pass http://healthsync_backend;
    access_log off;
}

# Block direct access to sensitive paths
location ~ /\. {
    deny all;
}
}

```

5.2 Enable and Test

```

sudo ln -s /etc/nginx/sites-available/healthsync /etc/nginx/sites-enabled/
sudo nginx -t
sudo systemctl reload nginx

```

6. SSL/TLS Configuration

6.1 For Internal Hospital Network (Self-Signed)

```

openssl req -x509 -nodes -days 365 \
    -newkey rsa:2048 \
    -keyout /etc/ssl/private/healthsync.key \
    -out /etc/ssl/certs/healthsync.crt \
    -subj "/C=NG/ST=Lagos/O=Hospital/CN=healthsync.hospital.local"

```

6.2 For Public-Facing (Let's Encrypt)

```
sudo apt install certbot python3-certbot-nginx
sudo certbot --nginx -d healthsync.hospital.com
sudo certbot renew --dry-run
```

7. Post-Deployment Verification

7.1 Verification Checklist

Run in order after deployment:

```
# 1. Application starts without errors
sudo systemctl status healthsync

# 2. Database connectivity
curl -s https://healthsync.hospital.local/login | grep -q "Sign in" && echo "PASS"
|| echo "FAIL"

# 3. Run acceptance tests
cd /opt/healthsync && npx tsx scripts/verify.ts

# 4. Check security headers
curl -sI https://healthsync.hospital.local/ | grep -E "(X-Content-Type|X-Frame|Strict-Transport|Content-Security)"

# 5. Verify HTTPS redirect
curl -sI http://healthsync.hospital.local/ | grep "301"

# 6. Test login with admin account
# (manual – log in via browser and verify dashboard loads)

# 7. Verify JWT secret is production-grade
# The application will refuse to start if JWT_SECRET is weak in production mode
```

7.2 Smoke Tests

Test	Expected Result	Pass/Fail
Login page loads	200 OK with sign-in form	
Admin login works	Redirect to /dashboard	
Patient registration	MRN generated, audit logged	
Appointment booking	No double-booking allowed	
Patient portal	Scoped to logged-in patient	
Nurse cannot prescribe	403 error returned	
Audit log visible to admin only	Other roles see 403	

8. Backup and Recovery

8.1 Automated Database Backups

Create `/opt/healthsync/scripts/backup.sh` :

```
#!/bin/bash
BACKUP_DIR="/var/backups/healthsync"
TIMESTAMP=$(date +%Y%m%d_%H%M%S)
RETENTION_DAYS=30

mkdir -p "$BACKUP_DIR"

pg_dump -U healthsync_user -h localhost healthsync_db \
  | gzip > "$BACKUP_DIR/healthsync_${TIMESTAMP}.sql.gz"

# Encrypt backup (PHI protection)
gpg --symmetric --batch --passphrase-file /opt/healthsync/.backup-key \
  "$BACKUP_DIR/healthsync_${TIMESTAMP}.sql.gz"
rm "$BACKUP_DIR/healthsync_${TIMESTAMP}.sql.gz"

# Rotate old backups
find "$BACKUP_DIR" -name "*.gpg" -mtime +$RETENTION_DAYS -delete

echo "$(date): Backup completed - healthsync_${TIMESTAMP}.sql.gz.gpg" >>
/var/log/healthsync-backup.log
```

Add to crontab:

```
0 2 * * * /opt/healthsync/scripts/backup.sh
```

8.2 Recovery Procedure

```
# Decrypt backup
gpg --decrypt "$BACKUP_DIR/healthsync_YYYYMMDD_HHMMSS.sql.gz.gpg" \
  | gunzip | psql -U healthsync_user -h localhost healthsync_db
```

9. Monitoring and Logging

9.1 Application Logs

```
# View live logs
sudo journalctl -u healthsync -f

# View errors only
sudo journalctl -u healthsync -p err
```


9.2 Log Rotation

Create `/etc/logrotate.d/healthsync` :

```
/var/log/healthsync/*.log {
    daily
    missingok
    rotate 90
    compress
    delaycompress
    notifempty
}
```

9.3 Health Monitoring

Add a health check API route and monitor with your preferred tool (Uptime Kuma, Nagios, etc.):

```
# Simple uptime check
curl -sf https://healthsync.hospital.local/login > /dev/null || alert "HealthSync
DOWN"
```

10. Scaling Considerations

10.1 Horizontal Scaling

HealthSync runs stateless (JWT sessions, no server-side session store). To scale:

1. Deploy multiple Node.js instances behind a load balancer
2. All instances share the same PostgreSQL database
3. Ensure `JWT_SECRET` is identical across instances
4. Use sticky sessions or ensure cookies reach any instance

10.2 Database Scaling

- Add read replicas for dashboard/reporting queries
- Consider connection pooling via PgBouncer for >50 concurrent users
- Enable PostgreSQL `pg_stat_statements` for slow query monitoring

11. Rollback Procedure

```
# 1. Stop the service
sudo systemctl stop healthsync

# 2. Restore previous version
cd /opt/healthsync
git checkout <previous-release-tag>
npm ci --production=false
npm run build

# 3. Rollback database (if migration was applied)
```

```
npx prisma migrate resolve --rolled-back <migration-name>

# 4. Restart
sudo systemctl start healthsync

# 5. Verify
curl -s https://healthsync.hospital.local/login | grep -q "Sign in"
```

12. Troubleshooting

Symptom	Likely Cause	Resolution
"JWT_SECRET is not set" on startup	Missing .env or variable	Verify .env exists and is readable by the service user
"JWT_SECRET must be at least 32 characters"	Weak secret in production	Generate a proper secret: <code>openssl rand -base64 48</code>
502 Bad Gateway	Node.js crashed	Check <code>journalctl -u healthsync</code> for errors
Database connection refused	PostgreSQL not running or wrong URL	Verify <code>DATABASE_URL</code> and PostgreSQL status
"EACCES" permission errors	Wrong file ownership	Run <code>chown -R healthsync:healthsync /opt/healthsync</code>
Slow page loads	Database not indexed	Run <code>ANALYZE</code> on PostgreSQL tables
Login fails for all users	JWT secret changed	All sessions are invalidated; users must re-login (expected)

End of Deployment Guide